SUJAL SHAH

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PROFESSIONAL SUMMARY

Final-year B.Tech student in AI & Machine Learning specializing in computer vision, model deployment, and cloud-scale ML infrastructure. Shipped production CNN classifiers (92% accuracy, 50K images) with FastAPI deployment pipelines, cutting inference latency 30%. Active R&D contributor on satellite-imagery encroachment detection and RAG-based document intelligence. Seeking an AI/ML Engineer role to build and scale data-driven ML systems end to end.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, R
AI / ML: TensorFlow, PyTorch, Scikit-learn, OpenCV, CNNs, Transfer Learning, Grad-CAM, SHAP, RAG, LLMs, Data Augmentation
Deployment & Cloud: FastAPI, REST APIs, Docker, AWS, Model Quantization, Batch Processing, Inference Pipelines
Web & Mobile: React.js, Node.js, Flutter, SQLite, JWT Authentication
Core Concepts: Deep Learning, Computer Vision, XAI, NLP, Satellite Imagery Analysis, Object Detection, RAG Systems
Tools: Git, Figma, Canva, PostGIS, QGIS

EXPERIENCE

AI / ML Intern · Endureva Innovations *Jan 2026 – Present*

- Build and deploy a CNN-based plant disease classifier (92% accuracy, 50K images, 38 classes), served via FastAPI with <200ms inference latency.
- Cut model inference time 30% through quantization and batch-processing optimizations, improving real-time throughput.
- Design end-to-end ML pipelines covering preprocessing, augmentation, training, evaluation, and production deployment.
- Integrate trained models into scalable REST APIs enabling real-time object detection and image classification in production.

PROJECTS

RAG AI Model — Civil Tender Document Intelligence *2025 – Present*

Python · LangChain · RAG · LLM · FastAPI · PDF Parsing · Vector Embeddings

- Built the PDF-parsing and vector-embedding ingestion pipeline powering a RAG platform that cuts tender analysis from hours to minutes.
- Implemented the contradiction-detector and synopsis-generator modules, serving results through FastAPI for end-to-end procurement intelligence.

Imagery-Based Unauthorised Construction Detection System *2025 – Present*

Python · YOLOv8 · Sentinel-1/2 · SAR · PostGIS · NDVI/NDBI · FastAPI

- Built the 3-model detection ensemble (YOLOv8 + SAR/NDVI-NDBI fusion) targeting ~95% accuracy and <4% false positive rate on pipeline encroachment detection.
- Used QGIS for geospatial analysis and validation, and served detection results through a FastAPI backend for the alert dashboard.

XAI-Care: Explainable Multimodal AI for Elderly Assistance *2025 – Present*

Python · TensorFlow · FastAPI · Grad-CAM · SHAP · Wearable Sensors

- Built the FastAPI backend delivering sub-second fall-detection alerts from fused wearable-sensor and camera data.
- Integrated Grad-CAM and SHAP explainability into the real-time inference pipeline for non-technical caregivers.

Water Survey — Flutter Field Application *2025*

Flutter · SQLite · REST API · Offline-First Architecture

- Designed the UX and form-validation flows for non-technical field staff capturing water connection data and meter readings offline.
- Built structured data export for municipal workflows and auto-sync on network availability.

AI Plant Disease Detection System *2026 – Present*

Python · TensorFlow · CNN · Flutter · Grad-CAM · Data Augmentation

- Designed the CNN architecture and data-augmentation strategy behind a 92%-accuracy, 38-class classifier trained on a 50K-image dataset.
- Deployed Grad-CAM–explained predictions via Flutter for rural, offline use.

Restaurant POS System (Offline-First) *2026 – Present*

Flutter · FastAPI · SQLite · JWT Authentication

- Built the backend for real-time order tracking and role-based access (Admin/Staff) on a fully offline POS system.
- Built the analytics engine surfacing daily revenue and peak-hour trends.

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning *Expected June 2026*
Sanjay Ghodawat University · CGPA: 8.15/10

12th Standard — Science (Computer Science) *2022*
Yashwantrao Chavan College of Science, Karad

ACHIEVEMENTS, CERTIFICATIONS & LANGUAGES

- Top 3 — Smart India Hackathon (College Round): Sign Language Learning App for English & Gujarati Sign Language conversion.
- Organized Prabal Hackathon (GDG SGU): Led the Creative Team for a 40-hour hackathon with 600+ participants across multiple colleges.
- Accenture Hackathon: Built an AI-powered fashion retail recommendation system under time-constrained conditions.
- Certifications: AWS Cloud Practitioner · Software Engineering Simulation (Accenture) · Technology Apprenticeship (Accenture)
- Languages: English (Fluent) · Hindi (Fluent) · Marathi (Fluent) · German (Basic)